MMI 807 — DEVELOPMENT OF OPEN EDUCATION RESOURCES AND TOOLS

Module 5: Constructive Criticism and Peer Review

MSc in Mathematical Innovation • Open University of Kenya

1 MODULE OVERVIEW

Any educational resource, no matter how carefully designed, benefits from the critical scrutiny of a knowledgeable peer before reaching learners. Peer review is not merely quality assurance — it is the mechanism by which the OER community collectively improves its resources and maintains standards. This module develops both the theoretical understanding and the practical skill of constructive peer review, with a specific focus on the review of mathematical OER. You will learn to give feedback that genuinely improves resources, to receive feedback productively, and to distinguish between review that serves the resource and review that serves the reviewer's preferences.

LEARNING OUTCOMES — By the end of this module, you will be able to:

1. State the principles of constructive peer review and explain why each principle matters.
2. Describe the review framework from both the author's and reviewer's perspective.
3. Construct a structured, evidence-based peer review of a mathematics OER.
4. Distinguish between constructive feedback (that improves the resource) and non-constructive feedback (that reflects reviewer preference or frustration).

2 PRINCIPLES OF CONSTRUCTIVE PEER REVIEW

2.1 1.1 What Makes Peer Review "Constructive"?

Constructive feedback is feedback that genuinely helps the creator improve their work. It is characterised by:

1. **Specificity:** "The explanation of the chain rule in Section 2.3 assumes knowledge of composite functions that learners at this level may not have" is specific. "The explanations are confusing" is not.
2. **Evidence-based reasoning:** Reference the specific element you are commenting on, the criterion it fails to meet, and the evidence for your assessment.
3. **Actionable recommendations:** Every identified problem should be accompanied by a specific, implementable suggestion for improvement.
4. **Balance:** Acknowledge genuine strengths alongside weaknesses. A review that only identifies problems fails to help the author know what to preserve.

5. **Separation of the work from the person:** The review is about the OER, not the author's ability, intelligence, or effort.
6. **Proportionality:** Distinguish between critical issues (mathematical errors, major accessibility failures) and minor improvements (phrasing, formatting).
7. **Professional respect:** The tone is collaborative, not superior. The reviewer is a partner in improvement, not a judge.

THE CONSTRUCTIVE FEEDBACK TEST

Before submitting any review comment, ask three questions:

1. **Is this specific?** Can the author identify exactly what I am referring to?
2. **Is this actionable?** Can the author do something concrete with this feedback?
3. **Is this about the work?** Am I commenting on a quality of the resource, or a preference of mine?

If any answer is "no," revise the comment before submitting.

2.2 1.2 Non-Constructive Review: What to Avoid

"This is very confusing and poorly written."	Not specific; not actionable; addresses the author, not the work.
"I would have done this completely differently."	Expresses reviewer preference, not a quality criterion.
"The GeoGebra applet doesn't work."	True but incomplete: what specifically doesn't work? On which device/browser? What happens instead?
"This is excellent! I have no suggestions."	Falsely positive review fails the resource and the author.
"The mathematics is wrong."	Asserts but doesn't demonstrate; doesn't specify which mathematics or what the error is.
"I don't think this will work for learners."	Vague; what kind of learners? What evidence? What specifically won't work?

3 THE REVIEW FRAMEWORK

3.1 2.1 The Reviewer's Framework

A structured review of a mathematics OER should evaluate the resource against explicit criteria, organised by category. The following framework is adapted from academic peer review and OER-specific quality frameworks:

Mathematical Accuracy	Are all mathematical claims, definitions, worked examples, and answer keys correct? Are any common misconceptions inadvertently reinforced?
Pedagogical Design	Does the OER build conceptual understanding or only procedural skill? Is the sequencing appropriate for the target learners?
Clarity and Language	Is the language clear, appropriate, and accessible? Are technical terms defined? Is the level appropriate for the target audience?
Accessibility	Does the OER meet the accessibility criteria from Module 4? Are there barriers for learners with disabilities?
Inclusivity	Are problem contexts culturally relevant? Are diverse learners represented? Does the OER work in low-resource settings?
Technical Functionality	Does the interactive content work as intended? On which platforms was it tested?
Licence and Attribution	Is the licence clearly stated and appropriate? Are all third-party materials properly attributed?
Alignment to Stated Goals	Does the OER achieve what it claims in its learning outcomes? Is there evidence it would produce the intended learning?

3.2 2.2 The Author's Framework: Receiving and Responding to Review

Authors receiving peer review often feel defensive. This is a natural response — the OER represents significant effort and intellectual investment. Managing this response productively requires:

- **Read the review completely before responding:** Resist the urge to rebut the first critical comment before reading the full review.
- **Separate your reaction from your response:** Your emotional reaction to criticism is valid; your response to the reviewer should be professional.
- **Respond to every comment:** Accept, partially accept (with explanation), or politely decline with a reasoned argument. Never ignore a review comment.
- **Distinguish preference from principle:** A reviewer who says "I would have used a different approach" is expressing a preference; a reviewer who says "this definition of a function is mathematically incorrect because it does not exclude one-to-many mappings" is identifying a quality issue.
- **Use a revision table:** For each review comment, document: what was suggested, what you changed (or why you did not change it), and where in the document the change appears.

4 WRITING A STRUCTURED OER REVIEW

4.1 3.1 Review Structure

A complete OER peer review has the following structure:

OER PEER REVIEW STRUCTURE

Section 1: Overview (2–3 sentences): What is the OER? What is its stated purpose and target audience?

Section 2: Strengths (specific, evidence-based): What does the OER do well? Which specific elements are effective and why?

Section 3: Critical Issues (labelled by category and priority): What are the most significant problems? Each item must include: the specific location, the criterion it fails, the evidence, and a specific recommendation.

Section 4: Minor Suggestions: Lower-priority improvements (phrasing, formatting, additional examples).

Section 5: Recommendation: One of: (a) Accept as submitted; (b) Accept with minor revisions; (c) Major revisions required; (d) Decline with encouragement to resubmit.

5 LEARNING ACTIVITIES AND ASSIGNMENT

5.1 4.1 Forum Activity

Write a summary (300 words) of the key guidelines for constructive peer review. Then select an OER from Module 1 (your own identified opportunity area), review one specific aspect of it using the structured review framework above, and post your review as a GitHub Gist or repository file. Share the link. Give constructive criticism of at least three colleagues' reviews — comment specifically on whether their feedback is specific, actionable, and work-focused.

5.2 4.2 Module 5 Assignment

MODULE 5 ASSIGNMENT: CONSTRUCTIVE OER REVIEW

Part 1: Review an OER (600–900 words)
Select one open educational resource from your Module 1 opportunity area. Write a complete structured peer review using the five-section framework from Section 3.1. Your review must:

- Evaluate at least five of the eight review categories from Section 2.1.
- Include at least three specific, actionable recommendations for improvement.
- Clearly distinguish between critical issues and minor suggestions.
- Conclude with a clear recommendation (accept, revise, or decline).

Submit the review to the course GitHub repository and post the link on the forum.

Part 2: Peer Review of Reviews (on discussion forum)
Give constructive criticism of at least three colleagues' OER reviews posted on the forum. For each review you critique, comment specifically on: (a) whether the feedback is specific enough to be actionable; (b) whether the review correctly identifies mathematical accuracy issues; (c) whether the review addresses accessibility and inclusivity adequately.

6 KEY CONCEPTS AND TERMINOLOGY

Constructive Feedback	Feedback that is specific, actionable, evidence-based, and work-focused, enabling the creator to improve their resource.
Peer Review	A structured evaluation of a resource by a knowledgeable peer, applying explicit criteria to identify strengths and areas for improvement.
Revision Table	A document tracking each review comment received, the author's response, and the specific change made (or justification for not changing).
Review Criteria	The explicit standards against which a resource is evaluated; criteria make reviews objective, consistent, and useful.
GitHub Gist	A lightweight GitHub tool for sharing code snippets, documents, or review notes with a shareable URL.

MODULE 5 SUMMARY

- Constructive peer review is specific, actionable, evidence-based, balanced, proportionate, and work-focused. Each principle is necessary; together they define the difference between review that improves resources and review that expresses reviewer frustration.
- A structured review framework (mathematical accuracy, pedagogical design, clarity, accessibility, inclusivity, technical functionality, licence, alignment) produces consistent, comprehensive, and fair evaluations.
- Authors must respond to every review comment, distinguishing between quality issues (which require revision) and reviewer preferences (which require only acknowledgement).
- Peer review in open communities is conducted publicly: GitHub enables transparent review history that benefits the entire community.
- *A good reviewer is the most valuable collaborator an OER developer can have. The goal is not to find fault — it is to help good work become better work.*